

Species-Aware Review: L-Lysine Biosynthesis via the Succinylated Diaminopimelate Pathway in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556)
Pathway bucket: KEGG ppu00300 "Lysine biosynthesis" / KEGG module **M00016** (Lysine biosynthesis, succinyl-DAP pathway, aspartate $\Rightarrow$ lysine) **Module area:** amino_acid_metabolism

1. Executive summary

The dedicated **succinylated diaminopimelate (succinyl-DAP) route** to L-lysine is **fully satisfiable** in *P. putida* KT2440. Every reaction from (S)-tetrahydrodipicolinate through meso-DAP to L-lysine — DapA $\rightarrow$ DapB $\rightarrow$ DapD $\rightarrow$ (DapC/transaminase) $\rightarrow$ DapE $\rightarrow$ DapF $\rightarrow$ LysA — maps to at least one candidate gene, and the two shared upstream precursor steps (aspartate kinase, aspartate-semialdehyde dehydrogenase) are also present. KT2440 is a **lysine prototroph and standard metabolic chassis** whose genome-scale models encode complete lysine biosynthesis

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;
re-annotated genome
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so the pathway is expected to be functional in vivo.

However, the candidate list is **noisy in ways that matter for curation**:

- **Paralog redundancy** exists at three steps: three DapA/K01714 genes, two LysA/K01586 genes, two DapF/K01778 genes. Sequence identity to *E. coli* references cleanly identifies the **primary ortholog** in each case (DapA = PP_1237; LysA = PP_2077; DapF = PP_5228) and marks the others as divergent paralogs / likely KO over-propagation.

- **The transamination step (step 4)** is covered by two mechanistically distinct candidates. **PP_1588 (*dapC*, K14267)** carries the DapC-specific InterPro/TIGRFAM signatures (IPR019878 "Succinyldiaminopimelate transaminase, DapC, beta/gammaproteobacteria"; TIGR03538) — **domain-level evidence that it is a genuine, lineage-appropriate DapC**, so it is the preferred step-4 gene. The **argD-type** enzyme PP_0372 (*aruC*, K00821, class-III PF00202) is a plausible bifunctional backup (it performs this reaction in *E. coli*). PP_1588's low pairwise identity to *E. coli* AspC/ArgD simply reflects that DapC is a distinct aminotransferase subfamily.
- **PP_3786 (*aspC*) → K10206 (DapL, EC 2.6.1.83)** is a **likely over-propagation**; the DapL bypass is not a gammaproteobacterial route.
- **Boundary contamination**: the candidate list includes threonine/methionine (hom/PP_1470, PP_0664) and peptidoglycan (murE/PP_1332, murF/PP_1333) genes that are **not** dedicated succinyl-DAP module members.

No pathway step is a genuine gap, and no new GO term or DapL/DapH module document is warranted for this organism.

2. Target-organism pathway definition

Included process. Cytoplasmic conversion of the aspartate-pathway intermediate L-aspartate-4-semialdehyde (+ pyruvate) into L-lysine through the DAP intermediates, using the **succinylase (acyl-protected) sub-variant** of the DAP pathway:

(S)-tetrahydrodipicolinate → N-succinyl-2-amino-6-oxopimelate → N-succinyl-LL-DAP → LL-DAP → meso-DAP → L-lysine.

meso-DAP is a **branch metabolite** simultaneously feeding peptidoglycan cross-linking (via MurE) and, after decarboxylation by LysA, cytoplasmic L-lysine.

Alternate names / database definitions. - KEGG: pathway map00300 "Lysine biosynthesis"; module **M00016** "Lysine biosynthesis, succinyl-DAP pathway, aspartate ⇒ lysine". M00016 definition: AK · ASADH · DapA(K01714) · DapB(K00215) · DapD(K00674) · [DapC K00821|K14267] · DapE(K01439) · DapF(K01778) · LysA(K01586|K12526). - MetaCyc: "L-lysine biosynthesis I" (DAP, succinylase variant). - UniProt PATHWAY tag: "L-lysine biosynthesis via DAP pathway; ... (succinylase route)".

Neighbouring pathways to keep separate. - **Aspartate-pathway trunk** (aspartate kinase, ASADH): shared precursor node feeding Lys **and** Thr/Met — keep outside the dedicated module boundary. - **Threonine/methionine branch** (homoserine dehydrogenase *hom*/PP_1470, PP_0664; EC 1.1.1.3). - **Peptidoglycan biosynthesis / cell-wall maps** (murE ppu00300-linked ligase, murF; consumers of meso-DAP, not synthesis steps). - **Arginine biosynthesis** (acetylornithine aminotransferase side of PP_0372). - Overview maps map01100 / map01110 / map01230 should not be used as the module boundary.

3. Expected step model (with target-organism call)

#	Step (enzyme)	EC	KO	Primary gene in KT2440	Other candidates	Module call
—	Aspartate kinase (precursor)	2.7.2.4	K00928	PP_4473 (aspK, PE1)	—	covered (shared precursor, outside module)
—	Aspartate-semialdehyde dehydrogenase (precursor)	1.2.1.11	K00133	PP_1989 (asd)	—	covered (shared precursor, outside module)
1	4-OH-tetrahydrodipicolinate synthase (DapA)	4.3.3.7	K01714	PP_1237 (56.8% to EcoDapA)	PP_2639 (31%), PP_2036 (26%)	covered
2	4-OH-tetrahydrodipicolinate reductase (DapB)	1.17.1.8	K00215	PP_4725	—	covered
3	THDP N-succinyltransferase (DapD)	2.3.1.117	K00674	PP_1530	—	covered
4	N-succinyl-DAP aminotransferase (DapC)	2.6.1.17	K14267 / K00821	PP_1588 (dapC/ K14267; DapC-specific TIGR03538/ IPR019878)	PP_0372 (argD-type, K00821, bifunctional backup)	covered
5	Succinyl-DAP desuccinylase (DapE)	3.5.1.18	K01439	PP_1525	—	covered
6	DAP epimerase (DapF)	5.1.1.7	K01778		PP_3790 (33%)	covered

#	Step (enzyme)	EC	KO	Primary gene in KT2440	Other candidates	Module call
				PP_5228 (57.6% to EcoDapF)		
7	DAP decarboxylase (LysA)	4.1.1.20	K01586	PP_2077 (lysA-I, 57.7%)	PP_5227 (lysA-II, 33%)	covered

All seven module steps → **covered**. No step is a gap or "not_expected".

4. Candidate genes and evidence

High-confidence, keep as primary enzyme for the step:

- **PP_4473 / aspK (Q88EI9)** — Aspartate kinase, EC 2.7.2.4, **PE1 (protein-level evidence)**, annotation score 4. The strongest-evidenced gene in the set; shared precursor (Lys/Thr/Met). *Curation: covered, mark as shared-precursor, outside module boundary.*
- **PP_1989 / asd (Q88LE4)** — ASADH, EC 1.2.1.11, PE3, score 4. Shared precursor.
- **PP_1237 / dapA (Q88NH2)** — DapA/HTPA synthase, EC 4.3.3.7. Curated UniProt keywords "Lysine biosynthesis / Diaminopimelate biosynthesis / Schiff base"; **56.8% identity to E. coli DapA** — the clear primary ortholog. UniProt PATHWAY = "(S)-tetrahydrodipicolinate from L-aspartate: step 3/4". *Note: local metadata mis-buckets it to ppu00261; belongs to lysine (ppu00300).*
- **PP_4725 / dapB (Q88DU4)** — DapB, EC 1.17.1.8, single-copy, PATHWAY step 4/4. Unambiguous.
- **PP_1530 / dapD (Q88MP1)** — DapD, EC 2.3.1.117, single-copy, PATHWAY "succinylase route step 1/3". Diagnostic of the succinylase variant.
- **PP_1525 / dapE (Q88MP5)** — DapE, EC 3.5.1.18, single-copy, PATHWAY "succinylase route step 3/3". Diagnostic of the succinylase variant.
- **PP_5228 / dapF (Q88CF3)** — DapF, EC 5.1.1.7, PF01678; **57.6% to E. coli DapF** and in an operon with PP_5227 → strongest DapF call.

- **PP_2077 / lysA-I (Q88L58)** — LysA/DAPDC, EC 4.1.1.20, PF02784; **57.7% to E. coli LysA** → primary DAP decarboxylase.

Transamination step (step 4) — covered, primary gene domain-supported:

- **PP_1588 / dapC (Q88MI3)** — KEGG K14267 (dedicated DapC), local EC 2.6.1.17. Although UniProt is thin (PE4 "Predicted", score 1, PF00155, no EC), **InterPro assigns the DapC-specific family IPR019878 "Succinyldiaminopimelate transaminase, DapC, beta/gammaproteobacteria" and NCBIfam TIGR03538 "succinyldiaminopimelate transaminase"** — a diagnostic, lineage-appropriate signature. This is the **preferred step-4 enzyme**. Its low pairwise identity to *E. coli* AspC/ArgD reflects that DapC is a distinct aminotransferase subfamily, not weak evidence.
- **PP_0372 / aruC (Q88QW2)** — KEGG K00821 = acetylornithine/N-succinyldiaminopimelate aminotransferase (EC 2.6.1.11 + **2.6.1.17**), class-III aminotransferase (PF00202/IPR005814, CDD OAT_like, PIRSF "4-aminobutyrate/lysine/ornithine transaminase"), 36.9% to *E. coli* ArgD. In *E. coli* the argD/ArgD-type enzyme physiologically performs succinyl-DAP transamination, so this is a **plausible bifunctional backup** for step 4; primarily an arginine-biosynthesis enzyme.

Likely over-propagation / lower-confidence paralogs:

- **PP_2639 / dapA (Q88JL0)** — K01714, curated lysine keywords, but only **31% to E. coli DapA**; divergent DHDPs-superfamily paralog. Possible functional back-up DapA or a specialised aldolase.
- **PP_2036 (Q88L99)** — K01714 but UniProt has only "Lyase" keyword, submission-name, score 1, **no EC**; **26% to E. coli DapA**, **28% to NanA/NAL**. Weakest DapA call — probably a non-canonical DHDPs/NAL-family enzyme, not a third lysine DapA.
- **PP_5227 / lysA-II (Q88CF4)** — K01586, EC 4.1.1.20, extra Pfam PF00278; **33% to E. coli LysA**. Divergent decarboxylase, but sits in an operon with a bona fide DapF (PP_5228), so plausibly a second functional LysA.
- **PP_3790 / dapF (Q88GD4)** — K01778, EC 5.1.1.7; **33% to E. coli DapF**; standalone. Divergent second epimerase.
- **PP_3786 / aspC (Q88GD8)** — KEGG K10206 = **DapL (LL-DAP aminotransferase, EC 2.6.1.83)**; local EC 2.6.1.1. UniProt: generic "Aminotransferase", PE4, no EC. InterPro places it only in the **broad IPR050881/PANTHER PTHR42832 "LL-diaminopimelate aminotransferase" family** and it **lacks the DapC signature** (no IPR019878/TIGR03538). **Likely over-propagation** — probably a generic aspartate/aromatic aminotransferase (see §5).

Not module members (boundary contamination in the candidate list):

- **PP_1470 / hom** and **PP_0664** — homoserine dehydrogenase, EC 1.1.1.3; UniProt PATHWAY = methionine/threonine. Belong to Thr/Met, not lysine.
- **PP_1332 / murE** — UDP-MurNAc-tripeptide:meso-DAP ligase, EC 6.3.2.13; **consumes** meso-DAP for peptidoglycan (downstream consumer, not a synthesis step).
- **PP_1333 / murF** — EC 6.3.2.10; peptidoglycan, unrelated to DAP synthesis.

5. Gaps, ambiguities, and likely over-annotations

1. **No genuine gaps.** All seven succinyl-DAP steps + both precursor steps are encoded. Module M00016 KO set is complete for *ppu*.
2. **Step-4 enzyme assignment (resolved by domain evidence).** PP_1588 carries the DapC-specific TIGR03538/IPR019878 signature and is the preferred dedicated DapC; PP_0372 (argD-type, K00821) is a plausible bifunctional backup. The step is **covered**; the residual uncertainty is only *in vivo* flux partitioning (which enzyme dominates), which is **not experimentally resolved in KT2440**. Note UniProt still assigns PP_1588 no EC, so the specific EC 2.6.1.17 rests on family signatures rather than direct assay.
3. **DapL over-propagation.** PP_3786→K10206 (DapL, EC 2.6.1.83) implies a THDP→LL-DAP shortcut. DapL is characterised in cyanobacteria, plants, Chlamydiae, Leptospira, Treponema and methanococci (PMIDs 16361515, 20418392, 25309529, 32274387), **not** in *Pseudomonas*/gammaproteobacteria, and KT2440 already has the full succinylase route. Treat as **likely over-propagation**; PP_3786 is more probably a generic aspartate/aromatic aminotransferase (local aspC/EC 2.6.1.1).
4. **DapA over-propagation.** Three genes carry K01714/EC 4.3.3.7 but only PP_1237 is a high-identity ortholog; PP_2639 and especially PP_2036 are divergent DHDPS-superfamily members that may not be lysine DapA.
5. **Broad/bifunctional EC mappings.** PP_0372 carries EC 2.6.1.11 + 2.6.1.13 (+2.6.1.17 via KEGG) — genuinely bifunctional (arginine + lysine); do not treat its lysine role as exclusive.
6. **Local bucket inconsistencies.** Several core lysine genes (PP_1237, PP_2036, PP_2639 dapA; PP_1989 asd; PP_4473 aspK; PP_4725 dapB) are tagged primary bucket **ppu00261** in the metadata but functionally belong to ppu00300/M00016; murF is tagged ppu01502. These are database-partition artifacts, not biology.

6. Module and GO-curation recommendations

Per-step status:

Step	Status	Note
DapA (1)	covered	primary PP_1237; PP_2639/PP_2036 candidate_uncertain paralogs
DapB (2)	covered	PP_4725, unambiguous
DapD (3)	covered	PP_1530, diagnostic of succinylase variant
DapC/ transaminase (4)	covered	primary PP_1588 (DapC-specific TIGR03538/IPR019878); PP_0372 (K00821) bifunctional backup; no direct assay in KT2440
DapE (5)	covered	PP_1525, diagnostic of succinylase variant
DapF (6)	covered	primary PP_5228; PP_3790 candidate_uncertain
LysA (7)	covered	primary PP_2077; PP_5227 second copy (operon with DapF)

Module boundary: the generic module boundary is broadly correct. Recommend: - **Exclude** hom/PP_1470, PP_0664 (Thr/Met) and murE/PP_1332, murF/PP_1333 (peptidoglycan) from the dedicated succinyl-DAP module. - **Keep** aspK/PP_4473 and asd/PP_1989 flagged as **shared precursor (outside module boundary)** — matches the generic scope note. - **Do not** open a DapL/aminotransferase-variant (DapH acetylase or DapL) module for this organism; **not_expected_in_target_taxon** for those variants.

GO / EC actions: - No new GO terms required — all activities map to existing GO/EC. - Flag PP_3786 GO/KO "LL-diaminopimelate aminotransferase (GO:0010285 / K10206)" for **review/removal** (over-propagation). - Flag PP_1588 EC 2.6.1.17 as **inferred by homology (IEA)**, not experimental. - **module_needs_revision:** minor — reassign the ppu00261-bucketed core genes (PP_1237, PP_2036, PP_2639, PP_1989, PP_4473, PP_4725) to the ppu00300/M00016 lysine module in local metadata.

7. Genes to promote to full fetch-gene review

Priority order:

1. **PP_3786 (aspC, Q88GD8)** — adjudicate/remove the DapL (K10206, EC 2.6.1.83) call vs. generic aspartate aminotransferase (EC 2.6.1.1); it lacks the DapC signature and DapL is not a *Pseudomonas* route. Highest-value correction.
2. **PP_1588 (dapC, Q88MI3)** — confirm and promote to a full DapC annotation (K14267, EC 2.6.1.17); supported by TIGR03538/IPR019878 but UniProt still gives no EC/name. Set as primary step-4 gene.
3. **PP_0372 (aruC, Q88QW2)** — clarify bifunctional acetylornithine (arginine) + succinyl-DAP (lysine backup) role (K00821, EC 2.6.1.11 + 2.6.1.17).
4. **PP_2036 (Q88L99)** and **PP_2639 (dapA, Q88JL0)** — determine whether either is a functional lysine DapA or a specialised/non-lysine DHDPS/NAL-family enzyme.
5. **PP_5227 (lysA-II) / PP_5228 (dapF)** — confirm the operon and whether PP_5227 is a second functional DAP decarboxylase.

8. Key references

- Nelson KE *et al.* (2002) Complete genome of *P. putida* KT2440. *Environ Microbiol* 4:799–808. (genome basis; via

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- Belda E *et al.* (2016) The revisited genome of *P. putida* KT2440... robust metabolic chassis. *Environ Microbiol* 18:3403–3424.

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(re-annotation; 1548 genes re-annotated).

- Yuan H *et al.* (2017) Pathway-consensus genome-scale metabolic reconstruction of *P. putida* KT2440. *Front Microbiol* /

P 28085902

(complete lysine biosynthesis in GSM).

- Hudson AO *et al.* (2006) LL-DAP aminotransferase (DapL) variant in plants/cyanobacteria.

P 16361515

(DapL taxonomic scope).

- Liu Y *et al.* (2010) Methanococci use the DapL pathway.

P 20418392

(DapL scope).

- Triassi AJ *et al.* (2014) DapL as a narrow-spectrum antibacterial target; DapL in ~13% of genomes (*Chlamydia*/*Leptospira*/*Treponema*).

P 25309529

(DapL not gammaproteobacterial).

- Peverelli MG *et al.* (2016) Dimerization of bacterial DAP decarboxylase essential for catalysis.

P 26921318

(LysA biology).

- Gokulan K *et al.* (2003) Crystal structure of *M. tuberculosis* LysA/DAPDC.

P 12637582

(LysA fold/mechanism).

Evidence provenance note

Direct *P. putida* KT2440 experimental evidence for individual step enzymes is limited (only aspK/PP_4473 is UniProt PE1). Satisfiability rests on: (i) complete KEGG M00016 KO assignments for *ppu*; (ii) UniProt curated pathway annotations; (iii) genome-scale metabolic models encoding lysine biosynthesis (strain-specific, strong); (iv) sequence-identity ortholog assignment to *E. coli* K-12 experimentally characterised enzymes (family-level transfer, strong for primary orthologs, weak for divergent paralogs); and (v) InterPro/TIGRFAM family signatures (decisive for the PP_1588 DapC call). The succinylase-variant assignment is firm because DapD (PP_1530) and DapE (PP_1525) are both present and are diagnostic of this sub-variant.

Limitations and open questions

- **No strain-specific enzyme assays.** With the exception of aspartate kinase, no KT2440 lysine-pathway enzyme has direct biochemical/genetic characterisation; all step assignments are homology/family inferences. Species transfer from *E. coli* is strong at the family level but does not establish in-vivo flux.
- **Step-4 flux partitioning unresolved.** Whether PP_1588 (dedicated DapC) or PP_0372 (argD-type) carries physiological succinyl-DAP transamination — or both — is unknown; a Δ PP_1588 / Δ PP_0372 auxotrophy test on minimal medium would resolve it.
- **Paralog functionality.** It is untested whether the divergent DapA (PP_2639, PP_2036), LysA (PP_5227) and DapF (PP_3790) paralogs are functional lysine enzymes, back-ups, or repurposed; single/double knockouts and complementation would clarify.
- **DapL call.** PP_3786 should be re-examined against a curated aspartate/aromatic-aminotransferase reference set; its EC 2.6.1.83 (K10206) is not supported by a DapL-specific signature and is very likely a mis-assignment.
- **No plasmid/second-copy caveats.** Analysis is chromosome-based (KT2440 is plasmid-free), so no additional lysine genes are expected elsewhere.